Supplementary Information 1: Detailed Analytical Proofs for "The Golden M_0 Tensor Mechanics and Quasi-Periodic 3-Torus ($\mathbb{T}^3_\phi$) Stability: Resolution of Resonance Collapse, Homological Information Preservation, and Proof of Lyapunov 0-Point Convergence via Golden Integer Lattices ($\mathbb{Z}[\phi]$)"

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Abstract

This supplementary document provides complete, self-contained mathematical derivations and step-by-step analytical proofs supporting Part 2 of the 2026 *Grand Unified Framework* series (*The Golden M_0 Tensor Mechanics and Quasi-Periodic 3-Torus ($\mathbb{T}^3_\phi$) Stability*). Specifically, we present: (i) the algebraic number theory of the Golden Integer ring $\mathbb{Z}[\phi] = \mathbb{Z}[\frac{1+\sqrt{5}}{2}]$ and the Pisot–Vijayaraghavan Galois contraction inequality $|\bar{\phi}|^{2n} = \phi^{-2n} < 1$; (ii) explicit multi-dimensional Diophantine lower bounds on both the transverse solenoidal Golden lattice $\mathcal{K}_\phi$ and the cubic Pisot extension, together with Sobolev energy estimates for the KAM homological equation; (iii) Hilbert-space spectral self-adjointness, idempotence, and exponential Lyapunov convergence $\Sigma(\boldsymbol{u}(t)) := \frac{1}{2}\mathrm{dist}^2_{\mathcal{H}}(\boldsymbol{u}(t), \mathcal{E}_0) \leq \Sigma(\boldsymbol{u}_0)e^{-2\alpha_\phi t} \to 0$; (iv) de Rham cochain-complex commutativity $\mathrm{d} \circ \boldsymbol{\mathcal{M}}_0^* = \boldsymbol{\mathcal{M}}_0^* \circ \mathrm{d}$ and Hodge–de Rham preservation of the first homology group $H_1(\mathbb{T}^3_\phi; \mathbb{Z}) \cong \mathbb{Z}^3$; and (v) the exterior flatness ($\nabla \times \boldsymbol{A}_{\mathrm{AB}} = \boldsymbol{0}$), quantized Golden holonomy ($\Delta\theta_k = 2\pi\phi^{-k}$), homological flux linkage across vortex cores, and non-trivial harmonic equilibrium witness $\boldsymbol{u}^* \in \mathcal{E}_0$ ($\|\boldsymbol{u}^*\|^2_{\mathcal{H}} = \frac{3}{2} > 0$). Each derivation is mapped 1:1 to `Supplementary_2_Formal_Verification_Lean4` and `Supplementary_3_CAS_Python_Proofs`.

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1 Algebraic Structure of the Golden Integer Ring $\mathbb{Z}[\phi]$ and Pisot Duality

Let $K = \mathbb{Q}(\sqrt{5}) = \{p + q\sqrt{5} : p, q \in \mathbb{Q}\}$ be the real quadratic number field of discriminant $\Delta_K = 5$. Because $5 \equiv 1 \pmod 4$, the ring of algebraic integers $\mathcal{O}_K$ is generated by the Golden Ratio ϕ:

$$\mathcal{O}_K = \mathbb{Z}[\phi] = \{a + b\phi : a, b \in \mathbb{Z}\}, \qquad \phi = \frac{1+\sqrt{5}}{2}, \qquad \bar{\phi} = \sigma(\phi) = \frac{1-\sqrt{5}}{2} = 1 - \phi = -\phi^{-1}. \quad (1)$$

Proposition 1.1 (Fundamental Identities, Norm Integrality, and Pisot Contraction in $\mathbb{Z}[\phi]$)**.** *For all elements $\xi = a + b\phi \in \mathbb{Z}[\phi]$ ($a, b \in \mathbb{Z}$) and all integers $n \geq 1$, the following exact algebraic identities hold (mechanized in `Theorem0_GoldenRing.lean` and verified symbolically in `CAS_Proof1_GoldenRing.py`):*

1. ***Minimal Polynomial and Unit Reciprocal:*** *$\phi^2 - \phi - 1 = 0$, $\bar{\phi}^2 - \bar{\phi} - 1 = 0$, $\phi + \bar{\phi} = 1$, and $\phi\bar{\phi} = -1$. Consequently, ϕ is a fundamental unit of $\mathbb{Z}[\phi]$ with inverse $\phi^{-1} = \phi - 1 = -\bar{\phi} \in \mathbb{Z}[\phi]$.*

2. ***Trace and Norm Integrality:*** *The field trace and norm map $\mathbb{Z}[\phi]$ into $\mathbb{Z}$:*

$$\mathrm{Tr}_{K/\mathbb{Q}}(\xi) = \xi + \bar{\xi} = 2a + b \in \mathbb{Z}, \qquad \mathrm{N}_{K/\mathbb{Q}}(\xi) = \xi\bar{\xi} = a^2 + ab - b^2 \in \mathbb{Z}. \quad (2)$$

   Moreover, $\mathrm{N}_{K/\mathbb{Q}}(\xi) = 0 \iff (a, b) = (0, 0)$, so every non-zero Golden integer $\xi \in \mathbb{Z}[\phi] \setminus \{0\}$ satisfies $|\mathrm{N}_{K/\mathbb{Q}}(\xi)| = |a^2 + ab - b^2| \geq 1$.

3. ***Pisot Geometric Series Normalization:*** *Since $|\bar{\phi}| = \phi^{-1} = \frac{\sqrt{5}-1}{2} \in (0, 1)$, we have $1 - \phi^{-2} = \phi^{-1}$ and the exact convergent geometric series:*

$$\sum_{n=0}^{\infty} (1 - \phi^{-2})|\bar{\phi}|^{2n} = \phi^{-1} \sum_{n=0}^{\infty} \phi^{-2n} = \frac{\phi^{-1}}{1 - \phi^{-2}} = 1. \quad (3)$$

Proof. (1) Direct calculation gives $\phi^2 = \frac{1+2\sqrt{5}+5}{4} = \frac{3+\sqrt{5}}{2} = 1 + \phi$, and $\phi\bar{\phi} = \frac{1-5}{4} = -1$. Thus $\phi(\phi - 1) = 1$, so $\phi^{-1} = \phi - 1 = -\bar{\phi}$.

(2) Expanding $\xi\bar{\xi} = (a + b\phi)(a + b\bar{\phi}) = a^2 + ab(\phi + \bar{\phi}) + b^2(\phi\bar{\phi}) = a^2 + ab - b^2 \in \mathbb{Z}$. Completing the square in $\mathbb{Q}$ gives $4(a^2 + ab - b^2) = (2a + b)^2 - 5b^2$. Since 5 is not a square in $\mathbb{Q}$, $(2a + b)^2 - 5b^2 = 0$ if and only if $b = 0$ and $2a + b = 0$, i.e., $a = b = 0$. Because $a^2 + ab - b^2$ is a non-zero rational integer whenever $(a, b) \neq (0, 0)$, its absolute value is at least 1.

(3) From $\phi^2 = \phi + 1$, dividing by ϕ^2 gives $1 = \phi^{-1} + \phi^{-2}$, so $1 - \phi^{-2} = \phi^{-1} \in (0, 1)$. Since $0 < \phi^{-2} = \frac{3-\sqrt{5}}{2} \approx 0.381966 < 1$, the geometric series $\sum_{n=0}^{\infty} \phi^{-2n}$ converges absolutely to $(1 - \phi^{-2})^{-1} = \phi$, establishing Equation (3). □

2 Multi-Dimensional Diophantine Lower Bounds and KAM Sobolev Estimates

We now provide the complete analytical proof of the Diophantine lower bounds (Equation (9) and Appendix A of the main text) and Theorem 3.1 (KAM Homological Equation Well-Posedness), corresponding to `Theorem1_Diophantine.lean` and `CAS_Proof2_Diophantine_Bound.py`.

Theorem 2.1 (Explicit Diophantine Lower Bounds on the Golden Lattice)**.** *Let $|\boldsymbol{k}| := |k_1| + |k_2| + |k_3|$ for $\boldsymbol{k} = (k_1, k_2, k_3) \in \mathbb{Z}^3 \setminus \{\mathbf{0}\}$.*

1. ***Quadratic Golden Foliation ($\mathcal{K}_\phi$):*** *For the pure Golden power frequency vector $\boldsymbol{\omega}_\phi = \omega_0(1, \phi^{-1}, \phi^{-2})^\mathsf{T} = \omega_0(1, \phi - 1, 2 - \phi)^\mathsf{T}$ ($\omega_0 > 0$), any wavevector $\boldsymbol{k} \in \mathbb{Z}^3$ satisfies $\boldsymbol{k} \cdot \boldsymbol{\omega}_\phi = \omega_0(A_{\boldsymbol{k}} + B_{\boldsymbol{k}}\phi)$, where $A_{\boldsymbol{k}} := k_1 - k_2 + 2k_3 \in \mathbb{Z}$ and $B_{\boldsymbol{k}} := k_2 - k_3 \in \mathbb{Z}$. On the transverse non-degenerate lattice $\mathcal{K}_\phi := \{\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\mathbf{0}\} : (A_{\boldsymbol{k}}, B_{\boldsymbol{k}}) \neq (0, 0)\} = \mathbb{Z}^3 \setminus \mathbb{Z}(1, -1, -1)$:*

$$|\boldsymbol{k} \cdot \boldsymbol{\omega}_\phi| = \omega_0 |A_{\boldsymbol{k}} + B_{\boldsymbol{k}}\phi| \geq \frac{\omega_0}{|A_{\boldsymbol{k}} + B_{\boldsymbol{k}}\bar{\phi}|} \geq \frac{\omega_0}{2\phi\,|\boldsymbol{k}|}, \qquad \forall\, \boldsymbol{k} \in \mathcal{K}_\phi. \quad (4)$$

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2. ***Full 3D Algebraic Integer Lattice ($\mathbb{Z}^3 \setminus \{\mathbf{0}\}$):*** *Let $(1, \alpha_1, \alpha_2)$ be a $\mathbb{Q}$-linearly independent basis of algebraic integers in a totally real cubic field $F/\mathbb{Q}$ of degree $d = 3$ with conjugate embeddings $\sigma_2, \sigma_3 : F \hookrightarrow \mathbb{R}$, and set $M_F := \max_{j\in\{2,3\}} \max(1, |\sigma_j(\alpha_1)|, |\sigma_j(\alpha_2)|) \geq 1$. Then for every $\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\mathbf{0}\}$:*

$$|\boldsymbol{k} \cdot \boldsymbol{\omega}_\phi| = \omega_0 |k_1 + k_2\alpha_1 + k_3\alpha_2| \geq \frac{\gamma_\phi}{|\boldsymbol{k}|^2}, \qquad \gamma_\phi := \frac{\omega_0}{M_F^2} > 0. \tag{5}$$

Proof. (1) Using $\phi^{-1} = \phi - 1$ and $\phi^{-2} = 2 - \phi$, we compute

$$k_1 + k_2\phi^{-1} + k_3\phi^{-2} = k_1 + k_2(\phi - 1) + k_3(2 - \phi) = (k_1 - k_2 + 2k_3) + (k_2 - k_3)\phi = A_{\boldsymbol{k}} + B_{\boldsymbol{k}}\phi \in \mathbb{Z}[\phi].$$

Notice that $A_{\boldsymbol{k}} = B_{\boldsymbol{k}} = 0 \iff k_2 = k_3$ and $k_1 = -k_2$, i.e., $\boldsymbol{k} \in \mathbb{Z}(1, -1, -1)$. For any $\boldsymbol{k} \in \mathcal{K}_\phi$, $\xi_{\boldsymbol{k}} := A_{\boldsymbol{k}} + B_{\boldsymbol{k}}\phi$ is a non-zero element of $\mathbb{Z}[\phi]$, so Proposition 1.1(2) gives $|\xi_{\boldsymbol{k}}||\bar{\xi}_{\boldsymbol{k}}| = |\mathrm{N}_{K/\mathbb{Q}}(\xi_{\boldsymbol{k}})| \geq 1$. By the triangle inequality, $|\bar{\xi}_{\boldsymbol{k}}| = |A_{\boldsymbol{k}} + B_{\boldsymbol{k}}\bar{\phi}| \leq |k_1| + (1 + |\bar{\phi}|)|k_2| + (2 + |\bar{\phi}|)|k_3| \leq (2 + \phi^{-1})|\boldsymbol{k}| = (\phi + 1)|\boldsymbol{k}| = \phi^2|\boldsymbol{k}| \leq 2\phi|\boldsymbol{k}|$. Dividing $|\xi_{\boldsymbol{k}}\bar{\xi}_{\boldsymbol{k}}| \geq 1$ by $|\bar{\xi}_{\boldsymbol{k}}|$ yields Estimate (4).

(2) Since $(1, \alpha_1, \alpha_2)$ are $\mathbb{Q}$-linearly independent algebraic integers in $\mathcal{O}_F$, $\zeta_{\boldsymbol{k}} := k_1 + k_2\alpha_1 + k_3\alpha_2 \in \mathcal{O}_F \setminus \{0\}$ for all $\boldsymbol{k} \in \mathbb{Z}^3 \setminus \{\mathbf{0}\}$. Thus its field norm $\mathrm{N}_{F/\mathbb{Q}}(\zeta_{\boldsymbol{k}}) = \zeta_{\boldsymbol{k}}\sigma_2(\zeta_{\boldsymbol{k}})\sigma_3(\zeta_{\boldsymbol{k}})$ is a non-zero rational integer, so $|\zeta_{\boldsymbol{k}}||\sigma_2(\zeta_{\boldsymbol{k}})||\sigma_3(\zeta_{\boldsymbol{k}})| \geq 1$. Since $|\sigma_j(\zeta_{\boldsymbol{k}})| \leq M_F(|k_1| + |k_2| + |k_3|) = M_F|\boldsymbol{k}|$ for $j \in \{2, 3\}$, dividing by $|\sigma_2(\zeta_{\boldsymbol{k}})||\sigma_3(\zeta_{\boldsymbol{k}})| \leq M_F^2|\boldsymbol{k}|^2$ proves Estimate (5). □

Corollary 2.1 (Sobolev Well-Posedness of the KAM Homological Equation)**.** *Under the Diophantine bound $|\boldsymbol{k} \cdot \boldsymbol{\omega}_\phi| \geq \gamma_\phi|\boldsymbol{k}|^{-\tau}$ ($\tau \geq 1$), for every zero-mean solenoidal perturbation $F(\boldsymbol{x}) = \sum_{\boldsymbol{k}\in\mathbb{Z}^3\setminus\{\mathbf{0}\}} \hat{F}_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{x}} \in H^s_\sigma(\mathbb{T}^3_\phi)$ ($s \geq \tau$), the unique zero-mean solution $\Psi = (\boldsymbol{\omega}_\phi \cdot \nabla)^{-1}F$ has Fourier coefficients $\hat{\Psi}_{\boldsymbol{k}} = \frac{\hat{F}_{\boldsymbol{k}}}{i(\boldsymbol{k}\cdot\boldsymbol{\omega}_\phi)}$ and satisfies:*

$$\|\Psi\|^2_{H^{s-\tau}(\mathbb{T}^3_\phi)} = \sum_{\boldsymbol{k}\neq\mathbf{0}} (1 + |\boldsymbol{k}|^2)^{s-\tau} \frac{|\hat{F}_{\boldsymbol{k}}|^2}{|\boldsymbol{k} \cdot \boldsymbol{\omega}_\phi|^2} \leq \frac{1}{\gamma_\phi^2} \sum_{\boldsymbol{k}\neq\mathbf{0}} (1 + |\boldsymbol{k}|^2)^{s-\tau}(1 + |\boldsymbol{k}|^2)^\tau |\hat{F}_{\boldsymbol{k}}|^2 = \frac{1}{\gamma_\phi^2}\|F\|^2_{H^s(\mathbb{T}^3_\phi)}. \tag{6}$$

3 Hilbert-Space Spectral Decomposition and Lyapunov Exponential Decay

This section provides the expanded proofs for Section 2.4–2.5 and Section 4.1 (Theorem 4.1) of the main text, corresponding to `Theorem2_M0Project.lean`, `Theorem3_LyapunovDecay.lean`, and `CAS_Proof3_M0_Commutator.py`.

Let $\mathcal{H} := H^1_\sigma(\mathbb{T}^3_\phi)$ be equipped with the volume-normalized inner product $\langle \boldsymbol{u}, \boldsymbol{v} \rangle_{\mathcal{H}} = \frac{1}{V_\phi} \int_{\mathbb{T}^3_\phi} g^{\alpha\beta}_{(\phi)} \nabla_\alpha u_i \nabla_\beta v_i \, \mathrm{d}V_{g^{(\phi)}}$, where $g^{(\phi)}_{\mu\nu} = \mathrm{diag}(1, \phi^{-2}, \phi^{-4})$, $g^{\mu\nu}_{(\phi)} = \mathrm{diag}(1, \phi^2, \phi^4)$, and $V_\phi = \int_{\mathbb{T}^3_\phi} \mathrm{d}V_{g^{(\phi)}} = (2\pi)^3\phi^{-3}$. In Fourier space, for any $\boldsymbol{u}(\boldsymbol{x}) = \sum_{\boldsymbol{k}\in\mathbb{Z}^3\setminus\{\mathbf{0}\}} \hat{\boldsymbol{u}}_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{x}}$, Plancherel's theorem gives:

$$\|\boldsymbol{u}\|^2_{\mathcal{H}} = \sum_{\boldsymbol{k}\in\mathbb{Z}^3\setminus\{\mathbf{0}\}} |\boldsymbol{k}|^2_{g^{(\phi)}} |\hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2, \qquad |\boldsymbol{k}|^2_{g^{(\phi)}} := g^{\alpha\beta}_{(\phi)} k_\alpha k_\beta = k_1^2 + \phi^2 k_2^2 + \phi^4 k_3^2. \tag{7}$$

Proposition 3.1 (Self-Adjointness, Idempotence, and Orthogonal Splitting of $\boldsymbol{\mathcal{M}}_0$)**.** *In the asymptotic Pisot limit where $\widehat{(\mathcal{S}_\sigma \boldsymbol{u})}_{\boldsymbol{k}} = (1 - \mathcal{I}_{\mathrm{res}}(\boldsymbol{k}))\hat{\boldsymbol{u}}_{\boldsymbol{k}}$ with spectral indicator $\mathcal{I}_{\mathrm{res}}(\boldsymbol{k}) \in \{0, 1\}$, the Golden M_0 operator $\boldsymbol{\mathcal{M}}_0 = \sum_{n=0}^\infty (1 - \phi^{-2})\phi^{-2n}\mathbb{P}_{\mathrm{Kakeya}}\mathcal{S}_\sigma^{n+1}$ acts on $\mathcal{H}$ as the Fourier multiplier $\widehat{(\boldsymbol{\mathcal{M}}_0\boldsymbol{u})}_{\boldsymbol{k}} = m_0(\boldsymbol{k})\hat{\boldsymbol{u}}_{\boldsymbol{k}}$ with $m_0(\boldsymbol{k}) = 1 - \mathcal{I}_{\mathrm{res}}(\boldsymbol{k}) \in \{0, 1\}$. Consequently:*

1. *$\boldsymbol{\mathcal{M}}_0^* = \boldsymbol{\mathcal{M}}_0$, $\boldsymbol{\mathcal{M}}_0^2 = \boldsymbol{\mathcal{M}}_0$, and $(\boldsymbol{I} - \boldsymbol{\mathcal{M}}_0)^2 = \boldsymbol{I} - \boldsymbol{\mathcal{M}}_0$;*
2. *For every $\boldsymbol{u} \in \mathcal{H}$, $\boldsymbol{u}_0 := \boldsymbol{\mathcal{M}}_0\boldsymbol{u} \in \mathcal{E}_0 = \ker(\boldsymbol{I} - \boldsymbol{\mathcal{M}}_0)$ and $\boldsymbol{u}_\perp := (\boldsymbol{I} - \boldsymbol{\mathcal{M}}_0)\boldsymbol{u} \in \mathcal{E}_0^\perp$ are strictly orthogonal in $\mathcal{H}$ ($\langle \boldsymbol{u}_0, \boldsymbol{u}_\perp \rangle_{\mathcal{H}} = 0$), satisfying Pythagoras' theorem $\|\boldsymbol{u}\|^2_{\mathcal{H}} = \|\boldsymbol{u}_0\|^2_{\mathcal{H}} + \|\boldsymbol{u}_\perp\|^2_{\mathcal{H}}$;*

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3. *The Lyapunov distance potential* $\Sigma(\boldsymbol{u}) = \frac{1}{2}\inf_{v_0\in\mathcal{E}_0}\|\boldsymbol{u}-v_0\|_{\mathcal{H}}^2 = \frac{1}{2}\|(\boldsymbol{I}-\boldsymbol{\mathcal{M}}_0)\boldsymbol{u}\|_{\mathcal{H}}^2$ *satisfies* $\Sigma(\boldsymbol{u}) \geq 0$ *and* $\Sigma(\boldsymbol{u}) = 0 \iff \boldsymbol{u} \in \mathcal{E}_0$.

Proof. Since $\mathcal{I}_{\text{res}}(\boldsymbol{k}) \in \{0,1\}$, we have $(1-\mathcal{I}_{\text{res}}(\boldsymbol{k}))^{n+1} = 1-\mathcal{I}_{\text{res}}(\boldsymbol{k})$ for all $n \geq 0$. Summing over $n \geq 0$ via Equation (3) gives $m_0(\boldsymbol{k}) = (1-\mathcal{I}_{\text{res}}(\boldsymbol{k}))\sum_{n=0}^{\infty}(1-\phi^{-2})\phi^{-2n} = 1-\mathcal{I}_{\text{res}}(\boldsymbol{k}) \in \{0,1\}$. Because $m_0(\boldsymbol{k})$ is real-valued and satisfies $m_0(\boldsymbol{k})^2 = m_0(\boldsymbol{k})$ and $m_0(\boldsymbol{k})(1-m_0(\boldsymbol{k})) = 0$, Plancherel's formula (7) immediately implies $\boldsymbol{\mathcal{M}}_0^* = \boldsymbol{\mathcal{M}}_0$, $\boldsymbol{\mathcal{M}}_0^2 = \boldsymbol{\mathcal{M}}_0$, and $\langle \boldsymbol{u}_0, \boldsymbol{u}_\perp\rangle_{\mathcal{H}} = \sum_{\boldsymbol{k}\neq\boldsymbol{0}}|\boldsymbol{k}|_{g^{(\phi)}}^2 m_0(\boldsymbol{k})(1-m_0(\boldsymbol{k}))|\hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2 = 0$. For any $v_0 \in \mathcal{E}_0$, since $\boldsymbol{u}_0 - v_0 \in \mathcal{E}_0 \perp \boldsymbol{u}_\perp$, $\|\boldsymbol{u}-v_0\|_{\mathcal{H}}^2 = \|\boldsymbol{u}_\perp\|_{\mathcal{H}}^2 + \|\boldsymbol{u}_0-v_0\|_{\mathcal{H}}^2 \geq \|\boldsymbol{u}_\perp\|_{\mathcal{H}}^2$, with equality at $v_0 = \boldsymbol{u}_0 = \boldsymbol{\mathcal{M}}_0\boldsymbol{u}$. □

Proposition 3.2 (Detailed Proof of Theorem 4.1: Coercive Grönwall Decay of $\Sigma(\boldsymbol{u}(t))$)**.** *For any solution* $\boldsymbol{u}(t)$ *of the* M_0*-regularized momentum equation (14) of the main text on* $\mathbb{T}_\phi^3$*, the orthogonal component* $\boldsymbol{u}_\perp(t) = (\boldsymbol{I}-\boldsymbol{\mathcal{M}}_0)\boldsymbol{u}(t) \in \mathcal{E}_0^\perp$ *satisfies:*

$$\frac{\mathrm{d}}{\mathrm{d}t}\Sigma(\boldsymbol{u}(t)) + \nu\left(\frac{1}{2}-\phi^{-2}\right)\|\nabla_{g^{(\phi)}}\boldsymbol{u}_\perp(t)\|_{\mathcal{H}}^2 \leq -2\alpha_\phi\Sigma(\boldsymbol{u}(t)) \quad\Longrightarrow\quad \Sigma(\boldsymbol{u}(t)) \leq \Sigma(\boldsymbol{u}_0)e^{-2\alpha_\phi t}, \tag{8}$$

where $\frac{1}{2}-\phi^{-2} = \frac{\sqrt{5}-2}{2} \approx 0.118034 > 0$ *and* $\alpha_\phi = \nu\lambda_1(\mathbb{T}_\phi^3)\phi^{-2} + (\gamma_M - C_{\Lambda,M}) > 0$.

Proof. Applying $\boldsymbol{Q}_0 = \boldsymbol{I}-\boldsymbol{\mathcal{M}}_0$ to the evolution PDE and taking the $\mathcal{H}$-inner product with $\boldsymbol{u}_\perp$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\Sigma(\boldsymbol{u}) = -\nu\|\nabla_{g^{(\phi)}}\boldsymbol{u}_\perp\|_{\mathcal{H}}^2 - \gamma_M\|\boldsymbol{u}_\perp\|_{\mathcal{H}}^2 + \frac{1}{V_\phi}\langle \boldsymbol{Q}_0(\mathbb{P}_\sigma[(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}] - \mathcal{F}_{\text{TZT}}), \Delta_{g^{(\phi)}}\boldsymbol{u}_\perp\rangle_{L^2}.$$

Using the Young viscous lock from Part 1, the nonlinear inner product on $\mathcal{E}_0^\perp$ is bounded by $\frac{\nu}{2}\|\nabla_{g^{(\phi)}}\boldsymbol{u}_\perp\|_{\mathcal{H}}^2 + C_{\Lambda,M}\|\boldsymbol{u}_\perp\|_{\mathcal{H}}^2$. Moving this to the right-hand side leaves $-\frac{\nu}{2}\|\nabla_{g^{(\phi)}}\boldsymbol{u}_\perp\|_{\mathcal{H}}^2$. Since $\phi^2 = \phi+1 > 2$, we have $\phi^{-2} = 2-\phi < \frac{1}{2}$, so $\frac{1}{2}-\phi^{-2} = \phi-\frac{3}{2} = \frac{\sqrt{5}-2}{2} > 0$. Splitting $-\frac{\nu}{2} = -\nu(\frac{1}{2}-\phi^{-2}) - \nu\phi^{-2}$ and applying Lemma A.2 of the main text ($\|\nabla_{g^{(\phi)}}\boldsymbol{u}_\perp\|_{\mathcal{H}}^2 \geq \lambda_1(\mathbb{T}_\phi^3)\|\boldsymbol{u}_\perp\|_{\mathcal{H}}^2$ since $|\boldsymbol{k}|_{g^{(\phi)}}^2 \geq |\boldsymbol{k}|^2 \geq \lambda_1(\mathbb{T}_\phi^3) = 1$ for all $\boldsymbol{k} \in \mathbb{Z}^3\setminus\{\boldsymbol{0}\}$) yields Inequality (8). Multiplying by $e^{2\alpha_\phi t}$ gives $\frac{\mathrm{d}}{\mathrm{d}t}(e^{2\alpha_\phi t}\Sigma(\boldsymbol{u}(t))) \leq 0$, completing the proof. □

4 Cochain-Complex Commutativity and Homological Period Preservation

This section provides the complete differential-form proof of Theorem 4.2 of the main text, corresponding to `Theorem4_HomologyPreserved.lean` and `CAS_Proof3_MO_Commutator.py`.

Proposition 4.1 (Explicit Fourier Verification of $\mathrm{d}\circ\boldsymbol{\mathcal{M}}_0^* = \boldsymbol{\mathcal{M}}_0^*\circ\mathrm{d}$ and Period Invariance)**.** *Let* $\alpha = \sum_{\mu=1}^3 \alpha_\mu(\boldsymbol{x})\,\mathrm{d}x^\mu \in \Omega^1(\mathbb{T}_\phi^3)$ *be a smooth 1-form on* $\mathbb{T}_\phi^3$ *with Fourier expansion* $\alpha_\mu(\boldsymbol{x}) = \hat{\alpha}_{\mu,\boldsymbol{0}} + \sum_{\boldsymbol{k}\in\mathbb{Z}^3\setminus\{\boldsymbol{0}\}}\hat{\alpha}_{\mu,\boldsymbol{k}}e^{i\boldsymbol{k}\cdot\boldsymbol{x}}$.

1. *For every* $\boldsymbol{k} \in \mathbb{Z}^3$*, the exterior derivative* $\mathrm{d}\alpha = \sum_{1\leq\mu<\nu\leq3}(\partial_\mu\alpha_\nu - \partial_\nu\alpha_\mu)\,\mathrm{d}x^\mu\wedge\mathrm{d}x^\nu$ *and the diagonal Fourier multiplier* $(\boldsymbol{\mathcal{M}}_0^*\alpha)_\mu(\boldsymbol{x}) = \sum_{\boldsymbol{k}\in\mathbb{Z}^3} m_0(\boldsymbol{k})\hat{\alpha}_{\mu,\boldsymbol{k}}e^{i\boldsymbol{k}\cdot\boldsymbol{x}}$ *commute identically:*

$$\big(\widehat{\mathrm{d}(\boldsymbol{\mathcal{M}}_0^*\alpha)}\big)_{\mu\nu,\boldsymbol{k}} = i\big(k_\mu m_0(\boldsymbol{k})\hat{\alpha}_{\nu,\boldsymbol{k}} - k_\nu m_0(\boldsymbol{k})\hat{\alpha}_{\mu,\boldsymbol{k}}\big) = m_0(\boldsymbol{k})\widehat{(\mathrm{d}\alpha)}_{\mu\nu,\boldsymbol{k}} = \big(\widehat{\boldsymbol{\mathcal{M}}_0^*(\mathrm{d}\alpha)}\big)_{\mu\nu,\boldsymbol{k}}. \tag{9}$$

2. *Along each fundamental 1-cycle* $\gamma_j : [0,2\pi] \to \mathbb{T}_\phi^3$ *(*$j \in \{1,2,3\}$*) given by* $\gamma_j(s) = s\boldsymbol{e}_j$*, every non-zero Fourier mode* $e^{i\boldsymbol{k}\cdot\gamma_j(s)}\mathrm{d}x^j = e^{ik_js}\mathrm{d}s$ *with* $k_j \neq 0$ *is exact (*$\mathrm{d}(\frac{1}{ik_j}e^{ik_js})$*) and integrates to zero over* $[0,2\pi]$*, while modes with* $k_j = 0$ *satisfy* $\boldsymbol{k} \in \mathcal{E}_0$ *(*$m_0(\boldsymbol{k}) = 1$*) or have zero j-th component by solenoidality* $k_j\hat{u}_{j,\boldsymbol{k}} = 0$*. Since* $m_0(\boldsymbol{0}) = 1$ *on the harmonic subspace* $\mathcal{H}_\Delta^1(\mathbb{T}_\phi^3) = \mathrm{span}\{\mathrm{d}x^1,\mathrm{d}x^2,\mathrm{d}x^3\}$*, the period integral is strictly invariant:*

$$\oint_{\gamma_j}\boldsymbol{\mathcal{M}}_0^*\alpha = 2\pi m_0(\boldsymbol{0})\hat{\alpha}_{j,\boldsymbol{0}} = 2\pi\hat{\alpha}_{j,\boldsymbol{0}} = \oint_{\gamma_j}\alpha, \qquad \forall\, j\in\{1,2,3\}. \tag{10}$$

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5 Force-Free Aharonov–Bohm Gauge Synchronization, Homological Flux Linkage, and Non-Trivial Harmonic Witness

Finally, we provide the explicit derivations for Theorem 5.1, Remark 5.1 (Moffatt homological flux linkage [6]), and Theorem 6 (Non-Trivial Satisfiability Audit) of the main text, corresponding to `Theorem5_ABHolonomy.lean`, `Theorem6_KAMAudit.lean`, and `CAS_Proof4_AB_Curvature.py`.

Proposition 5.1 (Force-Free Exterior Curvature, Quantized Holonomy, and Homological Flux Linkage). 1. ***Aharonov–Bohm Connection:*** *For the hydrodynamic gauge connection* $\boldsymbol{A}_{\mathrm{AB}}(\boldsymbol{x}) = \sum_{k=1}^{3} \phi^{-k} \nabla x_k = (\phi^{-1}, \phi^{-2}, \phi^{-3})^{\mathsf{T}}$ *on the multiply connected exterior domain* $\Omega_{\mathrm{ext}} = \mathbb{T}^3_\phi \setminus \bigcup_{a=1}^{M} K_a$*, every partial derivative* $\partial_i (\boldsymbol{A}_{\mathrm{AB}})_j = 0$ *vanishes identically on* Ω_{ext}*, so* $\mathcal{F}_{\mathrm{AB}} = \nabla \times \boldsymbol{A}_{\mathrm{AB}} \equiv \mathbf{0}$*, while integrating around the k-th fundamental cycle* $\gamma_k(s) = s\boldsymbol{e}_k$ *(*$s \in [0, 2\pi]$*) yields:*

$$\Delta\theta_k = \oint_{\gamma_k} \boldsymbol{A}_{\mathrm{AB}} \cdot \mathrm{d}\boldsymbol{x} = \int_0^{2\pi} \phi^{-k} \, \mathrm{d}x_k = 2\pi\phi^{-k}, \qquad k \in \{1, 2, 3\}, \tag{11}$$

satisfying the Golden recurrence relation $\Delta\theta_1 - \Delta\theta_2 = 2\pi(\phi^{-1} - \phi^{-2}) = 2\pi\phi^{-3} = \Delta\theta_3$*. Furthermore, for any two vortex core cross-sections* Σ_1, Σ_2 *bounding a solenoidal flux tube (*$\nabla \cdot \boldsymbol{\omega} = 0$*) along* $[\gamma_k] \in H_1(\mathbb{T}^3_\phi; \mathbb{Z})$*, Stokes' divergence theorem enforces exact homological flux conservation* $\iint_{\Sigma_1} \boldsymbol{\omega} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\Sigma + \iint_{\Sigma_2} \boldsymbol{\omega} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\Sigma = 0$ *[6].*

2. ***Non-Trivial Golden Equilibrium Witness (*$\boldsymbol{u}^* \in \mathcal{E}_0$*):*** *Define the smooth velocity field* $\boldsymbol{u}^* : \mathbb{T}^3_\phi \to \mathbb{R}^3$ *by*

$$\boldsymbol{u}^*(x_1, x_2, x_3) := \begin{pmatrix} \phi^{-1} \sin(x_2) \\ \phi^{-2} \sin(x_3) \\ \sin(x_1) \end{pmatrix}. \tag{12}$$

Then $\boldsymbol{u}^*$ *is* 2π*-periodic in each coordinate, has zero spatial mean* $\int_{\mathbb{T}^3_\phi} \boldsymbol{u}^* \, \mathrm{d}V_{g^{(\phi)}} = \mathbf{0}$*, is strictly divergence-free (*$\nabla \cdot \boldsymbol{u}^* = \phi^{-1}\partial_1 \sin(x_2) + \phi^{-2}\partial_2 \sin(x_3) + \partial_3 \sin(x_1) = 0$*), and is supported exclusively on the unit axial wavevectors* $\boldsymbol{k} \in \{(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)\} \subset \mathcal{K}_\phi$*, where* $|\boldsymbol{k} \cdot \boldsymbol{\omega}_\phi| \geq \omega_0 \phi^{-2} > 0$*. Thus* $\mathcal{I}_{\mathrm{res}}(\boldsymbol{k}) = 0 \implies \boldsymbol{\mathcal{M}}_0 \boldsymbol{u}^* = \boldsymbol{u}^* \in \mathcal{E}_0 \implies \Sigma(\boldsymbol{u}^*) = 0$*, while its* $\mathcal{H}$*-norm is strictly positive:*

$$\|\boldsymbol{u}^*\|^2_{\mathcal{H}} = \frac{1}{(2\pi)^3} \int_{[0,2\pi)^3} \left(\phi^2 \phi^{-2} \cos^2(x_2) + \phi^4 \phi^{-4} \cos^2(x_3) + \cos^2(x_1) \right) \mathrm{d}x_1 \mathrm{d}x_2 \mathrm{d}x_3 = \frac{1 + 1 + 1}{2} = \frac{3}{2} > 0, \tag{13}$$

proving that the zero-point attractor $\mathcal{E}_0$ *is non-vacuous and supports non-trivial solenoidal circulation.*

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6 Verification Mapping to Lean 4 (Supplementary_2) and SymPy CAS (Supplementary_3)

Table 1 provides the complete cross-reference matrix linking the analytical propositions proved in this document, the theorems of the main manuscript (Part2_Paper_v1.5-100.tex), the mechanized Lean 4 formal verification modules (Supplementary_2_Formal_Verification_Lean4), and the SymPy CAS symbolic scripts (Supplementary_3_CAS_Python_Proofs).

Table 1: Complete cross-reference matrix linking analytical derivations in Supplementary_1, main text theorems (Part2_Paper_v1.5-100.tex), Lean 4 formal proofs (Supplementary_2), and SymPy CAS symbolic scripts (Supplementary_3).

Supplementary 1	**Main Text Ref.**	**Lean 4 File (Supplementary_2)**	**SymPy CAS File (Supplementary_3)**
Proposition 1.1	Sec. 2.2, Eqs. (3)–(6), Theorem 3.2	Theorem0_GoldenRing.lean	CAS_Proof1_GoldenRing.py
Theorem 2.1, Corollary 2.1	Sec. 3.1–3.2, Theorem 3.1, Lemma A.1	Theorem1_Diophantine.lean	CAS_Proof2_Diophantine_Bound.py
Proposition 3.1	Sec. 2.4–2.5, Eqs. (8)–(9)	Theorem2_MOProject.lean	CAS_Proof3_MO_Commutator.py
Proposition 3.2	Sec. 4.1, Theorem 4.1, Lemma A.2	Theorem3_LyapunovDecay.lean	CAS_Proof3_MO_Commutator.py
Proposition 4.1	Sec. 4.2, Theorem 4.2, Eqs. (18)–(19)	Theorem4_Homology Preserved.lean	CAS_Proof3_MO_Commutator.py
Prop. 5.1(1)	Sec. 5.1, Theorem 5.1, Remark 5.1	Theorem5_ABHolonomy.lean	CAS_Proof4_AB_Curvature.py
Prop. 5.1(2)	Sec. 5.2, Table 2 (Theorem 6)	Theorem6_KAMAudit.lean	CAS_Proof4_AB_Curvature.py

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Acknowledgments

The author expresses sincere gratitude to Researcher He Ra Shin for her insightful discussions and steadfast support throughout the course of this work. Her foundational inspiration contributed significantly to the theoretical formulation and formal realization of this research.

Declarations

Data and Code Availability. All mechanized Lean 4 formal verification suites, symbolic CAS scripts, numerical solvers, and benchmark datasets supporting the findings of this study are publicly archived via the canonical repository (`https://github.com/arkhenomia/arkheion`) and the dataset archive (`https://huggingface.co/datasets/arkhenomia/arkheion`).

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